

Name:

SOLUTIONS

Date:



Year 9 Mathematics

Test 5, 2015

Topics – Statistics

48 39 = _____ %

Total Time: 60 minutes
Total Reading: 5 minutes
Total Working: 55 minutes
Weighting: _____ % of the year.

SECTION 1: CALCULATOR FREE

Time: 23 minutes	Marks for Section 1: 28 19
Reading: 2 minutes	Equipment Allowed: Nil
Working: 21 minutes	

1. [2 marks: ½ each]

categorical data
mean

range
skewed

continuous data
symmetrical

discrete data
mode

Complete the following using words from the list above.

- a) The number of goals scored by each team in one round of a netball competition is an example of

DISCRETE DATA (½P) ✓

- b) If a statistical graph is not skewed it is said to be SYMMETRICAL.

(½P) ✓

- c) The height of each student in your class is an example of CONTINUOUS DATA

(½P) ✓

- d) The difference between the highest and lowest score is the RANGE.

(½P) ✓

2. [2 marks]

Explain why your weight is considered to be continuous data rather than discrete even though you record it as 52 kg, 55 kg, etc.

BECAUSE IT CAN VARY BETWEEN THE
SMALLEST AND LARGEST VALUES.

Rounding, 52kg contains all values
from 51.5kg's to just under 52.5kg's

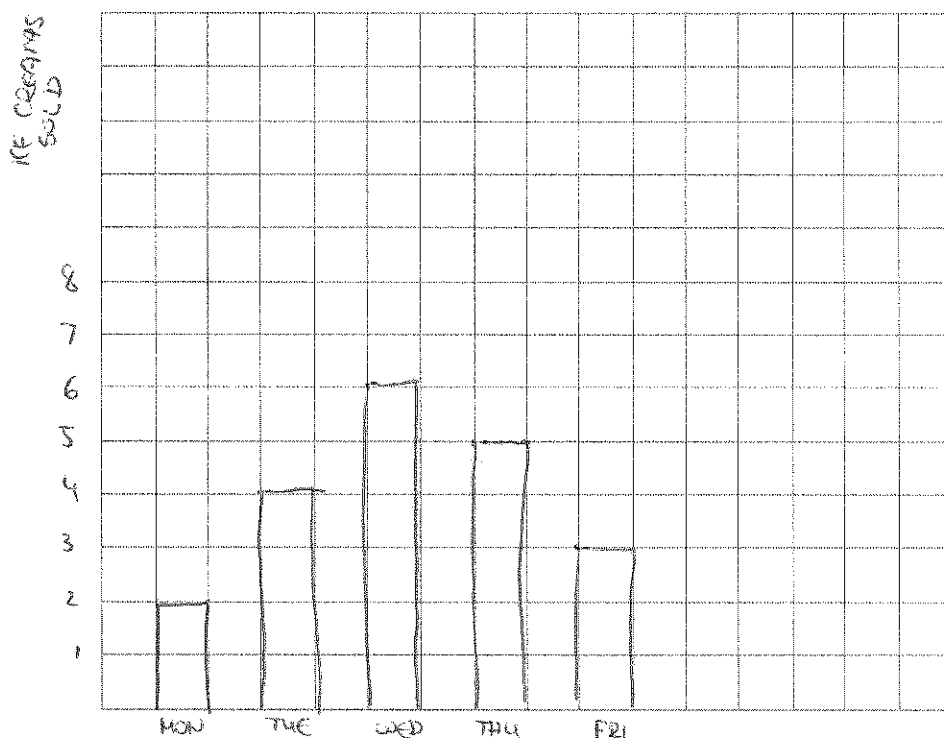
3. [5 mark: 2, 1, 1, 1]

The sale of ice creams from the school canteen was recorded for the past 15 days.

.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
Mon	Tues	Wed	Thurs	Fri

Ice cream sales

a) Display this information as a column graph.



b) When did the sale of ice creams peak?

✓ WED

c) What was the average number of ice creams?

✓ $(2+4+6+5+3) \div 5 = 4$

d) What day do you think was the coldest? Give reasons for your answer.

✓ MONDAY AS THE LEAST NUMBER OF ICECREAMS WERE SOLD.

4. [1 mark]

Abbey scores the following scores in 6 hands of a card game.

200, 180, 160, 140, 100 and 40.

Find the median of her scores.

$$\frac{160 + 140}{2} = 150 \checkmark$$

5. [2 marks]

Five friends earn the following marks on a quiz out of 10 marks:

5, 5, 6, 9 and 10.

Find the mean of their marks.

$$\frac{(5+5+6+9+10)}{5} = 7 \checkmark$$

6. [1 mark] ~~seven~~

Ellie plays ~~eight~~ games of netball and scores the following number of goals:

8, 3, 6, 7, 12, 9 and 14.

What is the range of her scores?

$$14 - 3 = 11 \checkmark$$

7. [1 mark]

Taylah records the number of minutes of exercise she does each day over two weeks.

	Mon	Tue	Wed	Thur	Fri	Sat	Sun
Week 1	20	45	30	20	60	45	60
Week 2	45	20	45	30	20	45	35

Find the mode of her times.

$$45 \checkmark$$

9. [3 marks: 1, 1, 1]

Mitchell's scores in 20 shots at a dartboard.

Stem	Leaf							
0	3	5	7	9	9			
1	0	4	7	8	9	9		
2	2	5	6	6	6	6	9	9
3								
4								
5								
6	0							

a) What is the mode of the scores?

✓ 26

b) Which score could be described as an outlier?

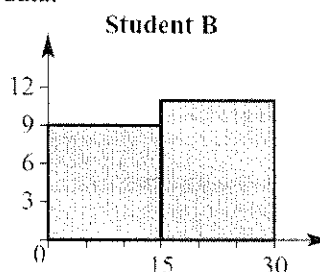
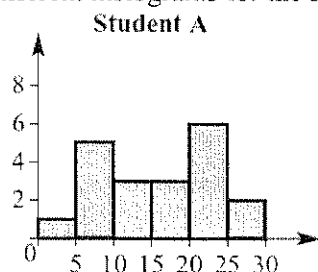
✓ 60

c) What effect would it have on the mean if the outlier were not included in the data? (You do not have to calculate the mean to answer this question.)

✓ MEAN WILL BE MORE ACCURATE AND SMALLER THEN THE MEAN WITH OUTLIER INCLUDED.

9. [2 marks: 1, 1]

Two students show different histograms for the same set of data.



a) Which histogram is more useful in helping to analyse the data? Justify your decision.

✓ HISTOGRAM OF STUDENT A AS IT IS MORE DETAILED IN SHOWING DATA SPREAD

b) What would you advise student B to do differently when constructing the histogram?

✓ MAKE SMALLER NUMBER SEGMENTS ON BOTH AXIS.

~ END OF TEST SECTION 1 ~

Name: _____

Date: _____

SECTION 2: CALCULATOR ASSUMED

Time:	37 minutes	Marks for Section 2:	20
Reading:	3 minutes	Equipment Allowed:	½ page notes (A4 one side), Scientific calculator
Working:	34 minutes		

10. [3 marks: 1, 1, 1]

Data was collected for a footy tipping competition one week, and the results are summarised below:

Score (x)	Frequency (f)	fx
6	4	24
7	9	63
8	10	80
9	7	63
$\Sigma f = 30$		$\Sigma fx = 230$

a) What is the mode of the scores?

✓ 8

b) What is the median of the scores?

✓ 8

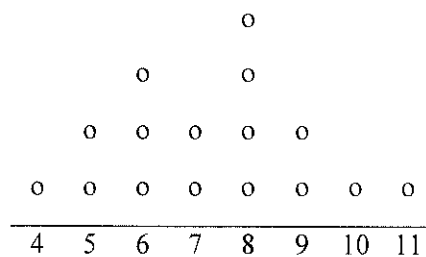
c) What is the mean of the scores (correct to one decimal place)?

$$\checkmark \frac{230}{30} = 7.6\bar{6}$$

$$= 7.7$$

11. [2 mark]

Heather records the ages of her 16 young cousins in the dot plot.



She realises that she has left out a cousin who is 7.

Which statistical measures (mean, median, mode, range) will change when she includes the extra score?

MEAN AND MEDIAN

✓

12. [2 marks]

The results on the same test for two classes are shown in the tables.

Ms Hypatia		Mr Kepler	
Score	Frequency	Score	Frequency
20	0	20	3
21	1	21	6
22	8	22	7
23	7	23	8
24	7	24	3
25	6	25	2

Explain why comparing the mode for each class would not give an accurate comparison of the two class's performances on the test.

MODE DOES NOT SAY ANYTHING ABOUT THE REST OF THE RESULTS, JUST WHAT IS THE MOST FREQUENT RESULT OCCURRING

mode one of lowest score in H.

14. [8 marks: 2, 3, 1, 2]

The following frequency table shows the distance thrown, in metres, by the students in the javelin competition at the Annual Athletics Carnival.

Distance (m)	Frequency
0-<10	4
10-<20	12
20-<30	24
30-<40	9
40-<50	1

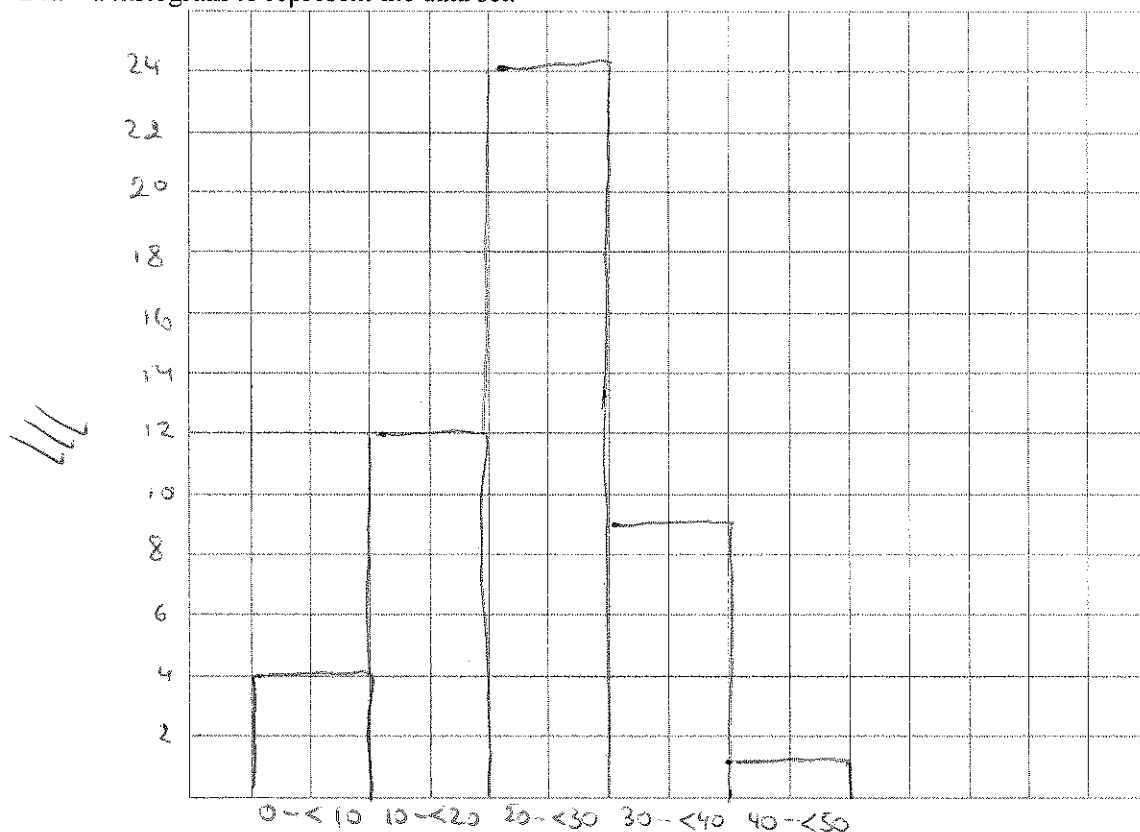
which one is it?

a) Calculate an estimate for the mean distance thrown by the competitors.

$$\bar{x} = \frac{5 \times 4 + 15 \times 12 + 25 \times 24 + 35 \times 9 + 45 \times 1}{4 + 12 + 24 + 9 + 1}$$

$$\bar{x} = 23.2$$

- b) Draw a histogram to represent the data set.



- c) Use the shape of the histogram to describe the data set.

HISTOGRAM IS FAIRLY SYMMETRICAL WITH DISTANCE PEAKING IN SEGMENT 20-30.

- d) 15 students were disqualified without recording a valid distance. Would it be reasonable to include these results? If you did record them what effect would they have on the mean and the median value?

IT'S NOT REASONABLE TO INCLUDE THEM AS MANY REASONS CAN INFLUENCE THE DISTANCE ACHIEVED, LIKE STEPPING OVER THE THROWING LINE.
MEAN AND MEDIAN ARE HIGHLY LIKELY TO CHANGE.

15. [5 marks]

The following lists are of heights, in cm, of 30 Year 9 students from each of NSW and Victoria recorded in 2011.

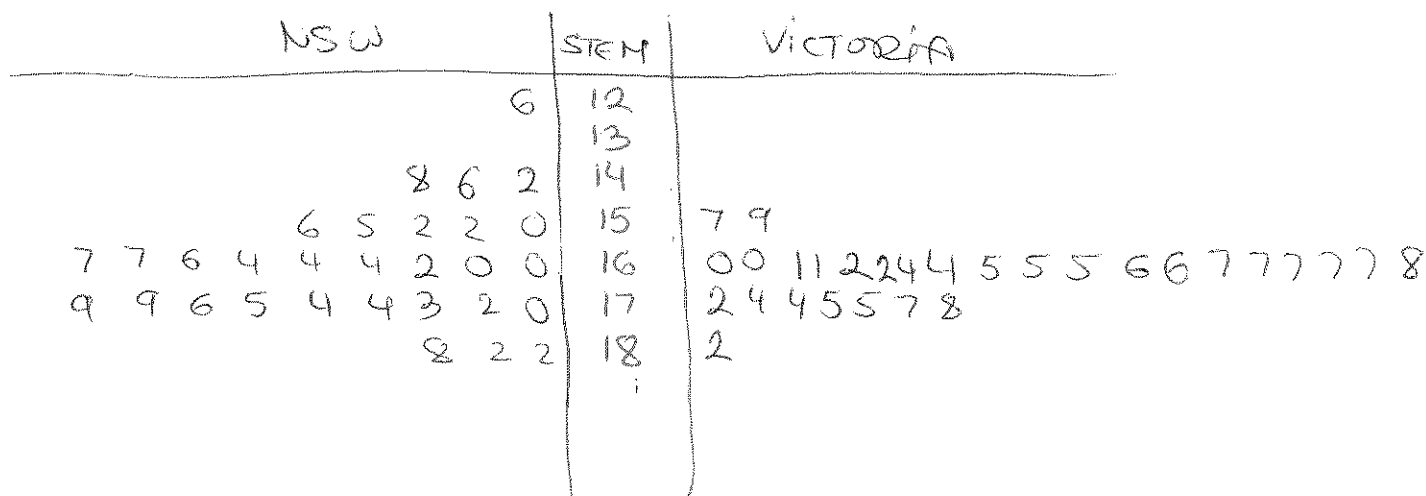
NSW students:

~~164~~ ~~160~~ ~~167~~ ~~164~~ ~~176~~ ~~156~~ ~~188~~ ~~164~~ ~~166~~ ~~174~~ ~~179~~ ~~167~~
~~142~~ ~~148~~ ~~175~~ ~~126~~ ~~155~~ ~~150~~ ~~173~~ ~~179~~ ~~146~~ ~~182~~ ~~152~~ ~~160~~
~~170~~ ~~172~~ ~~162~~ ~~174~~ ~~182~~ ~~152~~

Victoria students:

~~167~~ ~~164~~ ~~166~~ ~~186~~ ~~167~~ ~~165~~ ~~182~~ ~~161~~ ~~177~~ ~~157~~ ~~161~~ ~~162~~
~~165~~ ~~165~~ ~~172~~ ~~166~~ ~~167~~ ~~178~~ ~~160~~ ~~160~~ ~~167~~ ~~174~~ ~~175~~ ~~164~~
~~167~~ ~~175~~ ~~159~~ ~~168~~ ~~162~~ ~~174~~

Draw a back-to-back stem-and-leaf plot of the data.



~ END OF TEST SECTION 2 ~